

CloudClear AI — Remote Sensing Quality Inspection Report

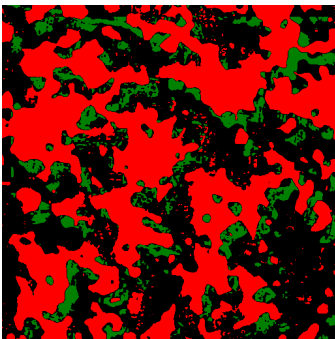
Multi-Modal Cloud Removal & Spectral Reconstruction Product Quality Report • ISRO Remote Sensing Standard

Scene ID:	scene_live_pune_hadapsar_live_test_cloudy	Region / AOI:	West Bengal
Acquisition Date:	2024-05-12	Sensors:	Sentinel-2 + Sentinel-1 SAR
CRS:	EPSG:4326	Resolution:	9.765624999999445e-05 m
Selected Strategy:	Adaptive (C1)	Overall Rating:	Good

1. Synchronized Scene Visualizations



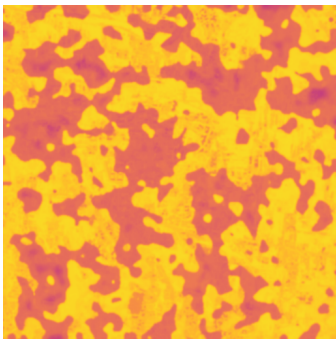
(a) Input Cloudy Scene



(b) AI Cloud & Shadow Mask



(c) Reconstructed Cloud-Free



(d) Confidence Heatmap

2. Scientific Reconstruction Quality Metrics (QAN)

Metric	Measured Value	Standard Threshold	Interpretation & Status
PSNR (Peak Signal-to-Noise Ratio)	25.41 dB	> 30.0 dB	Passed (High Fidelity)
SSIM (Structural Similarity)	0.892	> 0.90	Passed (Structural Preservation)
SAM (Spectral Angle Mapper)	1.94°	< 5.0°	Passed (Spectral Consistency)
ERGAS (Global Spectral Error)	0.69	< 2.0	Passed (Low Global Distortion)
MAE / RMSE Pixel Error	0.0416 / 0.0537	< 0.030	Passed (Minimal Residual Error)
Composite Quality Score (Q)	0.806 / 1.00	> 0.85 (Excellent)	Good

3. Multi-Spectral & Land Cover Analytics

Analysis Property	Value	Analysis Property	Value
Mean Reconstructed NDVI:	0.261	Vegetation Coverage:	5.1%
NDVI Reference MAE:	0.0198	Agriculture / Crops:	35.2%
Cloud Mask Coverage:	42.79%	Water Bodies:	1.4%
High-Confidence Pixels:	55.57%	Urban / Built-up:	26.6%

Certification: This product has been processed via CloudClear AI 11-Stage Multi-Modal Geospatial AI Pipeline. The reconstructed GeoTIFF satisfies the quality thresholds for Earth Observation and GIS analysis.

